

Roll No.....

MID SEMESTER EXAMINATION SEPTEMBER 2025

Name of the Program: B.Tech (Mechanical Engineering)

Semester: V

Name of the Course: Heat transfer

Course Code: TME 501

Time: 90 Minutes

Maximum Marks: 50

Note:

- (i) Answer **all the questions** by choosing **any one of the sub questions**.
- (ii) Each question carries 10 marks

Q1	(10 marks)	(CO1) & (CO3)
(a)	Derive the general three-dimensional heat conduction equation in Cartesian coordinates and mention its special cases.	
OR		
(b)	A cylindrical pipe of 60 mm ID and 70 mm OD carries hot fluid at 150°C. Outer surface is at 50°C. If $k = 20 \text{ W/m}\cdot\text{K}$, determine the heat loss per meter length .	
Q2	(10 marks)	(CO1) & (CO3)
(a)	Explain the concept of thermal resistance network for a composite wall. Derive the expression for the overall heat transfer coefficient .	
OR		
(b)	A cylindrical pipe of 60 mm ID and 70 mm OD carries hot fluid at 150°C. Outer surface is at 50°C. If $k = 20 \text{ W/m}\cdot\text{K}$, determine the heat loss per meter length .	
Q3	(10 marks)	(CO2) & (CO3)
(a)	Define fins and explain their importance. Derive the expression for temperature distribution and heat dissipation of a short fin with insulated tip .	
OR		
(b)	A straight fin 120 mm long, 8 mm thick, $k = 200 \text{ W/m}\cdot\text{K}$, base 180°C, ambient 30°C, $h = 25 \text{ W/m}^2\cdot\text{K}$. Calculate the heat dissipated (insulated tip).	
Q4	(10 marks)	(CO2) & (CO3)
(a)	Derive expressions for fin efficiency and fin effectiveness . Discuss their role in fin design.	
OR		
(b)	(b) A steel rod fin ($k = 45 \text{ W/m}\cdot\text{K}$), 10 mm diameter, 60 mm long, insulated at one end, exposed to air at 25°C ($h = 50 \text{ W/m}^2\cdot\text{K}$). Base temperature = 125°C. Find:(i) Fin efficiency, (ii) Heat dissipated.	
Q5	(10 marks)	(CO2) & (CO3)
(a)	Explain the mechanism of free convection and the role of Grashoff number with examples.	
OR		
(b)	A vertical plate 0.5 m high is maintained at 100°C in air at 30°C. Estimate the average heat transfer coefficient and the heat loss per meter width .	

